

ORF 309 – Midterm 1 Cheat Sheet

Chapters 1 – 2.3

1.3 Probability Axioms

Axioms: For probability measure P on $(\Omega, \mathcal{F})$:

1. $P(\Omega) = 1$
2. $P(A) \geq 0$ for all events A
3. (Countable additivity) If $A_1, A_2, \dots$ are disjoint:

$$P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i)$$

Consequences:

$$P(A^c) = 1 - P(A)$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$A \subseteq B \implies P(A) \leq P(B)$$

Inclusion–Exclusion:

$$P\left(\bigcup_{i=1}^n A_i\right) = \sum_i P(A_i) - \sum_{i < j} P(A_i \cap A_j) + \dots$$

Union Bound:

$$P\left(\bigcup_{i=1}^n A_i\right) \leq \sum_{i=1}^n P(A_i)$$

Uniform (equally likely): $|\Omega| < \infty$:

$$P(A) = \frac{|A|}{|\Omega|}$$

1.4 Counting

Permutations of k from n : $\frac{n!}{(n-k)!}$

Combinations: $\binom{n}{k} = \frac{n!}{k!(n-k)!}$

Multinomial: $\binom{n}{k_1, k_2, \dots, k_r} = \frac{n!}{k_1! k_2! \dots k_r!}$

1.5 Conditional Probability

$$P(A | B) = \frac{P(A \cap B)}{P(B)}, \quad P(B) > 0$$

Multiplication Rule:

$$P(A \cap B) = P(A | B) P(B)$$

Chain rule:

$$P(A_1 \cap \dots \cap A_n) = P(A_1) P(A_2 | A_1) P(A_3 | A_1 \cap A_2) \dots$$

Law of Total Probability: If $B_1, \dots, B_n$ partition Ω :

$$P(A) = \sum_{i=1}^n P(A | B_i) P(B_i)$$

Bayes' Rule:

$$P(B_i | A) = \frac{P(A | B_i) P(B_i)}{\sum_j P(A | B_j) P(B_j)}$$

1.6 Independence

A, B independent $\iff P(A \cap B) = P(A) P(B)$

Equivalently: $P(A | B) = P(A)$

Mutual independence: For *every* subcollection $S \subseteq \{1, \dots, n\}$:

$$P\left(\bigcap_{i \in S} A_i\right) = \prod_{i \in S} P(A_i)$$

Note: pairwise independence $\neq$ mutual independence.

1.7–1.8 Random Variables & Expectation

Indicator r.v.: $\mathbf{1}_A(\omega) = \begin{cases} 1 & \omega \in A \\ 0 & \omega \notin A \end{cases}$

$$P\{\mathbf{1}_A = 1\} = P\{A\}, \quad E[\mathbf{1}_A] = P\{A\}$$

Expectation (discrete):

$$E[X] = \sum_x x P\{X = x\}$$

Linearity of expectation (always holds):

$$E[aX + bY] = a E[X] + b E[Y]$$

Function of r.v.:

$$E[g(X)] = \sum_x g(x) P\{X = x\}$$

1.9 Independence & Conditioning of R.V.s

X, Y independent $\iff$ for all x, y :

$$P\{X = x, Y = y\} = P\{X = x\} P\{Y = y\}$$

If $X \perp Y$: $E[XY] = E[X] E[Y]$

Conditional expectation:

$$E[X | B] = \sum_x x P\{X = x | B\}$$

Tower property:

$$E[X] = \sum_i E[X | B_i] P(B_i)$$

2.1 Binomial Distribution

n independent Bernoulli(p) trials, $q = 1 - p$.

S_n = number of successes $\sim \text{Binom}(n, p)$:

$$P\{S_n = k\} = \binom{n}{k} p^k q^{n-k}, \quad k = 0, 1, \dots, n$$

$$E[S_n] = np$$

Binomial theorem: $(p + q)^n = \sum_{k=0}^n \binom{n}{k} p^k q^{n-k} = 1$

2.2 Geometric & Negative Binomial

Geometric: T_1 = trial of first success.

$T_1 \sim \text{Geom}(p)$:

$$P\{T_1 = n\} = q^{n-1} p, \quad n = 1, 2, 3, \dots$$

$$E[T_1] = \frac{1}{p}$$

Verification: $\sum_{n=1}^{\infty} q^{n-1} p = \frac{p}{1-q} = 1$

Murphy's law: $P\{T_1 < \infty\} = 1$ (success is guaranteed if $p > 0$).

Memoryless property:

$$P\{T_1 > m + n | T_1 > m\} = P\{T_1 > n\}$$

Negative Binomial: T_k = trial of k -th success.

$$P\{T_k = n\} = \binom{n-1}{k-1} p^k q^{n-k}, \quad n \geq k$$

$$E[T_k] = \frac{k}{p}$$

Inter-arrival times $T_i - T_{i-1}$ are i.i.d. $\text{Geom}(p)$.

Geometric series: $\sum_{k=0}^{\infty} r^k = \frac{1}{1-r}, \quad |r| < 1$

$$\sum_{k=0}^{\infty} k r^k = \frac{r}{(1-r)^2}$$

2.3 Law of Large Numbers

Variance:

$$\text{Var}(X) = E[(X - E[X])^2] = E[X^2] - (E[X])^2$$

Properties:

- $\text{Var}(X) \geq 0$; equals 0 iff X is constant a.s.
- $\text{Var}(aX) = a^2 \text{Var}(X)$
- $\text{Var}(X + c) = \text{Var}(X)$
- If $X \perp Y$: $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$

Bernoulli variance: $X \sim \text{Bern}(p)$: $\text{Var}(X) = p(1-p)$

Binomial variance: $S_n \sim \text{Binom}(n, p)$: $\text{Var}(S_n) = np(1-p)$

Chebyshev's inequality:

$$P\{|X - E[X]| \geq a\} \leq \frac{\text{Var}(X)}{a^2}$$

Law of Large Numbers:

$$E\left[\left(\frac{S_n}{n} - p\right)^2\right] = \frac{p(1-p)}{n} \rightarrow 0$$

For i.i.d. $X_1, X_2, \dots$ with $E[X_i] = \mu$:

$$\frac{X_1 + X_2 + \dots + X_n}{n} \rightarrow \mu \quad \text{as } n \rightarrow \infty$$